Chinese Sentiment Analysis Based on Knowledge Graph

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Abstract

With the rapid development of data mining technology, the use of knowledge graph for Chinese sentiment analysis has become a research hotspot in the field of natural language processing. This article aims to explore the combination of knowledge graph technology and Chinese sentiment analysis to enhance the depth and accuracy of sentiment analysis. It outlines the development trajectory and technical framework of knowledge graph, explains the theoretical foundation of sentiment analysis methods, and proposes relevant challenges and solutions based on the uniqueness of Chinese language. Building on the knowledge graph for sentiment analysis, this article designs an empirical research plan and reveals the practical application effects through rigorous data analysis. The research results show that knowledge graph can effectively improve the accuracy of Chinese sentiment analysis and the ability to handle complex texts. The article summarizes the innovations of the research and expectations for future work, providing theoretical basis and practical guidance for subsequent research.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Natural language processing; Information extraction.

Keywords

Knowledge graph, Chinese sentiment analysis, Natural language processing, Data mining, Affective computing, Technical framework

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In today's era of information explosion, massive amounts of Chinese text data continue to emerge, which contain rich emotional information. Sentiment analysis, as an important research direction in the field of natural language processing, aims to automatically identify and analyze emotional tendencies in texts. However, due to the complexity and ambiguity of the Chinese language, traditional sentiment analysis methods often face challenges such as word ambiguity and flexibility of grammatical structure when processing Chinese texts. In order to overcome these problems, Chinese sentiment analysis based on knowledge graphs has become an emerging method worth exploring.

As a structured and semantically rich knowledge representation method, knowledge graph can organize and express the relationships and attributes between entities, providing rich semantic information for sentiment analysis. By constructing a Chinese sentiment knowledge graph, elements such as sentiment words, sentiment events, and sentiment objects can be abstracted and expressed in the form of a graph, so that sentiment analysis is not limited to a single word-level analysis, but can better understand and infer the sentiment meaning in the text [1].

Chinese sentiment analysis based on knowledge graphs by introducing knowledge graphs, sentiment analysis systems can better understand emotional events in texts and can make inferences and inferences based on the rich semantic information in knowledge graphs, thereby improving the accuracy and robustness of sentiment analysis. This can not only overcome the challenges brought by the complexity of Chinese texts, but also give sentiment analysis stronger semantic understanding and reasoning capabilities. With the continuous development and improvement of knowledge graph technology, Chinese sentiment analysis based on knowledge graphs will surely play an important role in text sentiment understanding and application fields [2].

1 Overview of Knowledge Graph

The principles of knowledge graph technology mainly include entity recognition and linking, entity relationship extraction, knowledge representation and storage, knowledge reasoning and application, etc [3]. First, entity recognition and linking are the basis for knowledge graph construction. Its main task is to identify entities from text and data and link them to the corresponding entity nodes in the knowledge graph [4]. Secondly, entity relationship extraction refers to discovering the relationship between entities from text and data and representing them as edges in the knowledge graph [5]. In addition, knowledge representation and storage refers to storing and managing the extracted knowledge elements such as entities, relationships and attributes in the form of a graph database

to support efficient retrieval and query of knowledge [6]. Then, knowledge reasoning and application refers to automatic reasoning and application based on the knowledge graph to support application scenarios such as intelligent question answering, information recommendation, and knowledge discovery [7, 8].

In general, knowledge graphs automatically transform text, data, and knowledge into structured representations to support people's understanding and application of complex knowledge and information. In Chinese sentiment analysis based on knowledge graphs, these technical principles can help us extract sentiment information from massive Chinese texts and build sentiment knowledge graphs to support sentiment analysis and applications. Therefore, in-depth understanding and application of knowledge graph technical principles are of great significance to further promote the development of Chinese sentiment analysis.

2 Chinese-specific Challenges

When facing the technical research of Chinese sentiment analysis, specific language challenges have become an important factor affecting the accuracy and depth of analysis. The complexity of the Chinese language has always been a daunting task in the field of natural language processing (NLP). The following are some of the key challenges faced by Chinese sentiment analysis.

2.1 Language Specificity

Compared with English, the grammatical structure of Chinese is more flexible and the relationship between sentence components is more complex, which makes it more difficult to identify emotional expressions in sentences and requires more sophisticated dependency analysis. Chinese words often have multiple meanings, and the same emotion can be expressed by multiple words, which requires the model to have a higher level of semantic understanding ability. At the same time, Chinese idioms and colloquialisms also play an important role in sentiment analysis. They have rich meanings but are difficult to capture accurately.

2.2 Characteristics of Online Texts

Social media texts, such as microblogs and comments, are usually short in length, with a large amount of information but limited context, resulting in serious data sparsity problems. This sparsity will affect the training effect of the model and reduce the accuracy of sentiment classification. Online texts use a large number of online terms, slang, and emoticons. These expressions are highly informal and innovative, which poses a huge challenge to traditional dictionary-based sentiment analysis methods. At the same time, the same emoticon or Internet term may express different emotions in different contexts, which increases the complexity of sentiment analysis.

2.3 Influence of Cultural Background

Chinese expression habits are influenced by Chinese culture. Many emotional expressions are implicit and delicate, and need to be understood in combination with cultural background. For example, some derogatory words may be used to praise in certain situations. This use of irony and humor increases the difficulty of sentiment analysis. China is vast, and there are differences in language habits and cultural backgrounds in different regions. The same emotional expression may have different interpretations in different regions. This requires the sentiment analysis model to adapt to the language characteristics of different regions.

In summary, the Chinese specificity challenge is a complex challenge in the field of sentiment analysis that requires the integration of interdisciplinary knowledge. This study has achieved effective response to these challenges through technological innovation and the application of deep learning methods, and improved the accuracy and efficiency of Chinese sentiment analysis.

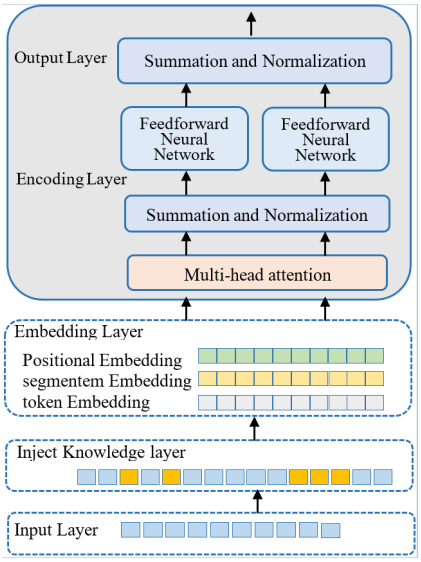

Figure 1: Comprehensive model of sentiment analysis based on knowledge graph.

3 Construction of Chinese Sentiment Analysis Model Based on Knowledge Graph

In the study of Chinese sentiment analysis, choosing a suitable classification model is crucial to improving the accuracy and efficiency of the analysis [9]. At present, classification models are mainly divided into machine learning methods, deep learning methods, sentiment dictionary-based and knowledge graph-based methods according to the technical scope adopted [10]. The overall structure of the comprehensive model based on knowledge graph proposed in this paper is shown in Figure 1. By integrating knowledge graph and deep learning technology, the accuracy of sentiment analysis is improved and the dependence of deep learning models on large-scale data is improved. The model mainly includes input layer, knowledge embedding layer, sentence representation layer, encoding layer and output layer.

3.1 Input Layer

In the input layer, simple preprocessing such as deduplication, sentence segmentation, and deletion of sentences with conflicting sentiments is sufficient. Sentence-level sentiment analysis assumes that the sentiment polarity of all aspects in a sentence is consistent.

Therefore, in the process of preprocessing the review text, sentences with conflicting sentiment polarities are deleted.

3.2 Inject Knowledge Layer

Inject Knowledge layer The main task of the Emotion Knowledge Graph is to Inject emotional knowledge. According to the emotional knowledge graph, the knowledge is integrated into the sentence. The emotional knowledge graph is used to complete the implicit feature words to form semantically complete explicit knowledge.

S is used to represent a sentence, as shown in Formula 1. A sentence is represented by a word w, w_i represents the i-th word, and n represents the length of the sentence.

$$s = \{w_1, w_2, \cdots\cdots, w_n\} \tag{1}$$

$$T = Inject\,(s,k) = \left\{w_1, w_2, \cdots, w_i\left\{(o_j, p_j)\cdots(o_k, p_k)\right\}, \cdots, w_n\right\} \tag{2}$$

Input a sentence and obtain the representation of the sentence through the knowledge graph, as shown in Formula 2. The words in the sentence $w_i$form the sentiment knowledge triple <w_i-o_j-p_j>...<w_i-o_k-p_k>, w_i where is the opinion word that appears in the sentence, o_j indicates that p_j the sentiment polarity of p_j the opinion word w_i in the aspect is o_j, o_k, p_k indicates that the sentiment polarity of the opinion word $w_i$in $O_k$the aspect is P_k, which reflects that the same opinion word may express different sentiment polarities when evaluating different aspects. The sentiment knowledge in the knowledge graph SAKG is embedded into the sentence to form a sentence representation rich in sentiment knowledge.

3.3 Embedding Layer

Embedding Layer uses BERT to encode the input sentence. After knowledge embedding, the sentence is vectorized and the corresponding symbol embedding (token embedding), segment embedding bedding and position embedding embedding) is constructed by summing up. Embedding is the corresponding representation of a word or Chinese character in a pre-trained dictionary (vocab), segment Embedding is used to divide sentence a and sentence b, and is generally used to predict the next sentence prediction) task, if it is a single sentence, only sentence a is used. Here, since the sentiment polarity classification is performed on a single sentence, only sentence a is used. position Embedding is the expression of the position information corresponding to each word.

3.4 Encoding Layer

The encoding layer of the model consists of a multi-head-Attention and layer normalization. Normalization) and a feed forward neural network are superimposed. Multi-head attention concatenates the self-attention outputs of multiple different single heads into one, and then reduces the dimension of the output through a fully connected layer. The multi-head attention mechanism calculates the input data, and the calculation formulas are shown in 3 and 4.

$$\mathrm{Att}\,(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_K}}\right)V \tag{3}$$

$$V_i = \mathrm{Linear}\,(W_i \mathrm{concat}\,(\mathrm{Att}_1, \mathrm{Att}_2, \cdots\cdots, \mathrm{Att}_n) + b_i) \tag{4}$$

Where Q=K=V, indicating that each word must calculate attention with all words in the same sentence. Each attention calculation includes calculating the correlation between words, normalizing the correlation, and obtaining the encoding of the target word by weighted summation of the correlation and the encoding of all words. Here, the softmax normalization method uses Relux (). $d_K$is the dimension of the K vector, and $\sqrt{d_K}$the scaling factor effectively controls the vanishing gradient.

3.5 Output Layer

Output Layer Layer) predicts the sentiment classification of the sentence. Sentiment classification determines the overall sentiment category of the sentence based on the sentiment features extracted by the hierarchical attention layer and the target aspect of the input. The sentiment polarity is predicted by the fully connected layer softmax() function, as shown in Formulas 5 and 6.

$$p_y(y|x) = \mathrm{softmax}\left(W_y + b_y\right) \tag{5}$$

$$\mathrm{Loss} = -\sum_{i=1}^{N} Q\,(i,j)\,\lg(P(i|j)) + \lambda||\theta||^2 \tag{6}$$

N is the size of the data set; $Q(i,j)$is 1 or 0 , indicating whether the result is correct; $P(i|j)$is the probability of the model predicting the sample; $\lambda||\theta||^2$is the regularization term.

4 Experimental Data and Results Analysis

4.1 Dataset Analysis

This paper adopts a knowledge graph-based method to combine text sentiment analysis with semantic networks to deeply explore the emotional color of text. The experimental design selects diversified data sources for sentiment tendency assessment. By crawling well-known online forums, social media microblogs, news comments, e-commerce platform product reviews, and film and television website reviews, etc. Table 1 describes the attributes and properties of each data source.

In terms of data processing details, taking the product review dataset of an e-commerce platform as an example, this study uses the method of automatic text segmentation and sentiment word matching to transform complex user evaluation information into structured data, thereby evaluating the user sentiment tendency of various products and providing companies with accurate customer feedback analysis. At the same time, the process of label verification and secondary confirmation is optimized to improve the data reliability of the film review sentiment corpus.

In addition, the relationship between the amount of data and the proportion of sentiment words in the data source description table is analyzed, revealing the potential bias and unique value of different data sources in sentiment analysis. This table provides a clear information framework and data benchmark for the entire sentiment analysis process, laying a data foundation for the empirical analysis of this study.

In general, this study not only focuses on the semantic analysis of text content, but also focuses on the deep mining of emotional color. With the support of knowledge graph, it broadens the vision of traditional sentiment analysis and improves the accuracy of analysis. Through in-depth exploration of different data sources,

Table 1: Data source description table.

Data Source Name	Internet Forum	Weibo sentiment	news comment	Product Reviews	Movie review sentiment	WeChat Official Account
The amount of data	12000	8500	9300	15700	4300	6800
Positive Valence Words	28.50%	26.30%	29.70%	30.20%	27.10%	25.60%
Negative evaluation words	26.20%	24.10%	21.60%	25.30%	29.40%	23.70%
Positive emotion words	15.30%	19.40%	17.50%	18.80%	16.70%	21.10%
Negative sentiment words	16.80%	18.90%	19.20%	14.60%	15.50%	17.40%
Advocate vocabulary	0.70%	1.20%	0.80%	1.10%	2.10%	0.90%
Level Vocabulary	12.50%	10.10%	11.20%	9.80%	9.20%	11.30%
Data Sources	User review collection	Weibo public data	News Site Comments	User evaluation information	Movie Review Section	Public account articles
Data processing method	Text preprocessing and denoising	Sentiment Annotation	Natural Language Processing	Text segmentation and sentiment word matching	Label verification and secondary confirmation	Topic extraction and sentiment analysis

the universality and richness of details of the research results are ensured, which has a positive role in promoting further research in the field of Chinese sentiment analysis.

4.2 Experimental Results Analysis

In order to analyze the Chinese sentiment analysis problem based on knowledge graph, the model constructed in this study analyzed several typical data sets and statistically analyzed the experimental results. The specific experimental process is as follows: First, extract text data from public data sources such as Weibo and news comments, generate training sets and test sets containing sentiment tags, and perform data preprocessing, including Chinese word segmentation, stop word removal, part-of-speech tagging and other steps; then, construct a sentiment analysis model based on knowledge graph. Compared with traditional models, the method that integrates knowledge graph is more accurate in understanding sentence connotation and capturing sentiment tendency.

Statistics show that the experimental results have significantly improved in multiple dimensions. Table 2 shows the statistical performance of different methods on Weibo sentiment, news comment count, and movie review data sets. In particular, in terms of positive emotion accuracy and negative emotion accuracy, the BERT model performed most prominently on the news review data set, with accuracy rates reaching 90.3% and 91.7% respectively. At the same time, it is advocated that detection accuracy and degree-level recognition accuracy are also key indicators for testing the performance of sentiment analysis models. From the statistical results, it can be seen that the BERT model also has excellent performance on these two indicators.

Each model also shows different performance advantages and disadvantages in the F1 score. As a comprehensive indicator of model accuracy and recall, the F1 score is an important reference for measuring the overall performance of the model. Comparing the experimental results, we found that the BERT model enhanced by the knowledge graph achieved the highest F1 score of 0.904. In comparison, the traditional dictionary analysis method, although relatively basic in this study, also showed good baseline performance.

Through in-depth comparative analysis of the above methods, it is verified that the Chinese sentiment analysis model combined with knowledge graph has better sentiment understanding ability. This provides experience reference for subsequent related research and promotes the depth and breadth of the application of knowledge graph in the field of Chinese sentiment analysis. At the same time, this study also provides a new research path for the future field of sentiment analysis, broadens the application scenarios of artificial intelligence in natural language processing, and has important theoretical significance and application value.

5 Conclusion

After studying and demonstrating Chinese sentiment analysis based on knowledge graph, the following conclusions can be drawn:

First, the application of knowledge graphs can effectively improve the accuracy and efficiency of Chinese sentiment analysis. By modeling the relationship between sentiment words and entities, we can better understand the sentiment expression in the text, thereby reducing ambiguity and improving the accuracy of sentiment analysis.

Secondly, the establishment and maintenance of knowledge graphs requires a large amount of domain knowledge and semantic information. When conducting Chinese sentiment analysis, it is necessary to continuously update and improve the content of the

Table 2: Experimental results statistics.

Method Name	data set	Positive review accuracy	Negative review accuracy	Positive sentiment accuracy	Negative sentiment accuracy	Claim detection accuracy	Degree level recognition accuracy	F1 score
Dictionary Analysis	Weibo comments	86.5%	84.2%	79.8%	81.3%	74.2%	67.8%	0.812
Naive Bayes	news comment	88.3%	85.7%	82.1%	83.5%	78.6%	71.4%	0.831
LSTM Network	Movie Review	91.2%	89.4%	87.5%	88.1%	82.9%	76.3%	0.869
CNN Model	Weibo sentiment	89.7%	87.9%	85.3%	86.0%	80.5%	73.9%	0.849
BERT Model	news comment	93.4%	92.1%	90.3%	91.7%	87.2%	82.6%	0.904

knowledge graph to ensure that it can cope with the ever-changing language expressions and social contexts.

In addition, the construction of knowledge graphs also faces some challenges, including the complexity of semantic understanding, the sparsity and incompleteness of data, etc. In practical applications, it is necessary to continuously improve the construction and maintenance algorithms of knowledge graphs to overcome these challenges.

Then, Chinese sentiment analysis technology based on knowledge graph is expected to be widely used in the fields of public opinion monitoring, text sentiment analysis, intelligent customer service, etc. With the continuous advancement of knowledge graph and artificial intelligence technology, we can expect this technology to play a greater role in the future.

Finally, in summary, Chinese sentiment analysis technology based on knowledge graph is of great significance in improving the accuracy of sentiment analysis, solving the problem of semantic understanding and promoting the application of artificial intelligence technology. Future research work needs to further improve the knowledge graph construction and application technology to promote the development of this field.

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